

Topic 7. COMPUTING AND NETWORKS (5G)

Life before the computer:

- ☞ *memory was something you lost with age*
- ☞ *an application was for employment*
- ☞ *a program was a TV show*
- ☞ *a web was a spider's house*
- ☞ *a virus was a flue*
- ☞ *a hard drive was a long trip on the road*

WATCH THE FOLLOWING VIDEO

https://www.youtube.com/watch?v=O5nskjZ_GoI

Early Computing: Crash Course Computer Science #1 (2017, 12 min)

Answer the following questions:

1. Why does the speaker start with the phrase "Hello, world!"?

2. According to the speaker, why are computers the lifeblood of today's world? What would happen if computers had to turn off?

3. What is the speaker's definition of a computer?

4. What is the first computer and how does it work?

5. What are other computing devices and their functions mentioned in the video?

6. What is the earliest documented use of the word ‘computer’? What did it mean?

7. What is Step Reckoner? Who invented it? What could it do and how?

8. What are three computing steps that followed a Step Reckoner?

9. What could the Difference Engine do?

10. What was Charles Babbage’s contribution to computing? Why is he considered the father of computing?

11. Why is Ada Lovelace considered the first programmer?

12. What was the challenge of 1890 census in the USA? How was the issue solved?

13. What is the connection between Herman Hollerith and IBM?

TOPIC RELATED VOCABULARY

1. Complete the sentences with the verbs from the box in the correct form.

crash drag freeze browse crop adjust click

- a) He _____ the Internet for hours looking for the right app and he still has not found one.
- b) The mouse can be used to _____ the file and move it into a new folder.
- c) The screen _____ and the cursor won't move.
- d) The employee's computer just _____ and he lost all his work.
- e) To edit a photo, use an appropriate app or program and _____, resize or _____ color and brightness as desired.
- f) To bookmark a page open the desired page on a web browser and _____ the bookmark icon in the top right hand corner.

2. The quote to this unit addresses the issue of how the meanings of the words changed with the advent of computing. Come up with the definition and sentences where these words: **crash, drag, freeze, browse** are used in their everyday, pre-computing meaning.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

- a) Having WiFi means you do not have to use cables to connect your computer ____the Internet.
- b) This new program scans your PC ____viruses.
- c) Everyone should be able to benefit ____the development of automated cars.
- d) I am not familiar____this program.
- e) Have you been in contact ____ the client today about the software upgrade?

The IT department will _____ the installation of the new program. It is designed to _____ issues with performance and monitor Internet usage. Nobody will be able to access the intranet while technicians work on the system and _____ any issues that arise. At the end of the upgrade they will _____ tests to ensure that everything is working correctly. When the results of the second test are the same and do not _____ the first, the work will be complete. The new program will not allow you to _____ your Internet usage. A warning message will _____ when you are close to doing so. After implementation we invite staff to _____ their feelings and feedback to us.

READING

Work in groups of 3. Each student reads an article on a current issue in computing. Retell to the other students in the group what your article is about. Discuss why it is an important issue. Check understanding of the vocabulary in the articles. Choose 2-3 words that are new/useful for you and share within your group.

1. What is 5G? An electrical engineer explains

Published: January 10, 2022 1.37pm GMT

Prasenjit Mitra, Professor of Information Sciences and Technology, Penn State, USA

5G stands for fifth-generation cellular network technology.

It's the technology that enables wireless communication – for example, from your cellular phone to a cell tower, which channels it to the internet. 5G is a network service provided by telecommunications carriers and is not the same thing as the 5 GHz band on your Wi-Fi router.

5G offers an order of magnitude – 10 times – more bandwidth than its predecessor, 4G. The greater bandwidth is possible because over and above low and medium frequency radio waves, 5G uses additional higher-frequency waves to encode and carry information.

Bandwidth is analogous to the width of a highway. The broader the highway, the more lanes it can have and the more cars it can carry at the same time. This makes 5G much faster and able to handle many more devices.



Most newer smartphones work with 5G networks

5G can deliver speeds of around 50 megabits per second, up to more than 1 gigabit per second. A gigabit per second connection allows you to download a high-definition movie in less than a minute. Does this mean no more bad cell connections in crowded places? The increased bandwidth will help, but just as increasing the number of lanes on highways does not always reduce traffic jams, as more people use the expanded highways, 5G is likely to carry a lot more traffic than 4G networks, so you still might not get a good connection sometimes.

In addition to connecting your phone and cellular-enabled laptop, 5G will be connecting many other devices ranging from photo frames to toasters as part of the Internet of Things revolution. So even though 5G can handle up to a million devices per square

kilometer, all that bandwidth could be quickly used up and require more – a future 5.5G with even more bandwidth.

Flavors of 5G

5G can use low-, mid- and high-band frequencies, each with advantages and disadvantages. Lower-frequency waves can travel farther but are slower. Using higher frequency waves means information can travel faster but these waves can only go limited distances. Higher-frequency 5G can achieve gigabit-per-second speeds, which promises to render ethernet and other wired connections obsolete in the future. Currently, however, the higher frequency comes at a higher cost and thus is deployed only where it's most needed: in crowded urban settings, stadiums, convention centers, airports and concert halls.

A type of 5G service, Ultra-Reliable and Low-Latency Communications, can be used where data needs to be transmitted without loss or interruption in service – for example, controlling drones in disaster areas. One day, after the technology is more robust, it could even be used for remote surgery.

2.Explainer: quantum computation and communication technology

Published: July 5, 2012 9.10pm BST

Howard Wiseman, Professor in Physics, Griffith University, Australia

What is a quantum technology?

Quantum mechanics is the branch of physics that explains the behaviour of matter and energy at the atomic scale. So does “quantum technology” just mean technology based on very small things? Is it just a cool-sounding synonym for nanotechnology?

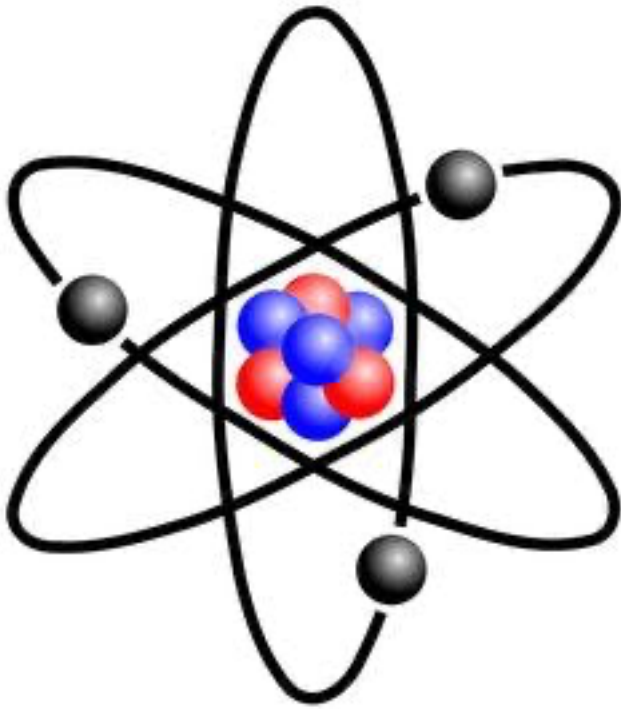
No! Quantum mechanics is certainly needed for designing a nanotechnology such as flash memory, but a quantum technology is something different.

It is a technology that makes use of the weirdness of quantum mechanics in a fundamental way - a technology whose operation makes no sense if you don't understand quantum mechanics.

In particular, quantum weirdness allows for impossibly fast computers and unhackable communication, which will be some of the defining technologies of the 21st century.

What's so weird about quantum mechanics?

In everyday life, we usually have a good sense of intuition regarding how the physical world will behave. Drop a glass and it will smash on the floor. Punch a concrete wall and your fist won't go through it.

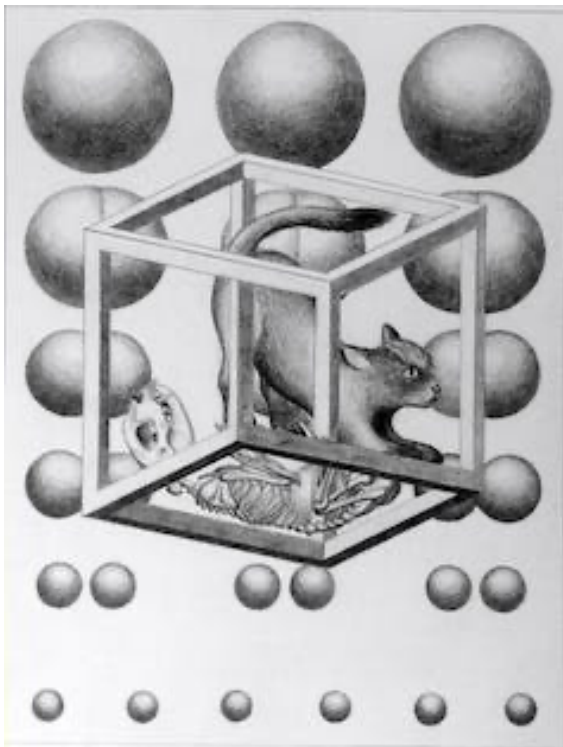


Wikimedia

But in the world of the ultra-small – atoms, electrons, even very dim light – none of the normal rules apply. Instead particles follow quantum rules that are quite baffling.

Heisenberg's Uncertainty Principle says that if you are sure how fast a particle is moving, you won't be able to predict where you would find it if you look for it. And if you are sure where the particle is, you won't be able to predict the value you'd get for its speed if you measure that instead.

Moreover, quantum theory says that this is not due to any limitations on our experimental abilities – the world itself is uncertain!



jieq

The really disturbing thing is that, according to quantum theory, there is nothing to stop this intrinsic uncertainty about microscopic matter growing into intrinsic uncertainty about everyday objects.

This is illustrated by the infamous Schrödinger's cat paradox, which says that a cat in a box could end up in a state where it's impossible to say whether it's alive or dead. It's not just that we are ignorant of whether the cat's alive or dead – until we open the box to look, it's neither.

Or both. Even quantum physicists disagree about how to describe it.

Ultrafast quantum computing

Quantum computing is Schrödinger's cat put to work. By building a computer out of quantum particles, it's possible to make the entire memory of the computer intrinsically uncertain, just like the health of the cat. This allows the computer to carry out astronomically many different computations in parallel.

The real trick is to bring all those computations together at the end to give the single answer you want. But if this could be done, a relatively small quantum computer could solve mathematical problems that couldn't be solved with all the present computers in the world running for a million years.



isamizdat

The first useful quantum algorithm – a piece of software for a quantum computer – that managed this trick was discovered in the 1990s.

The far greater challenge is to build the quantum computer hardware that will work reliably enough, and is big enough, to run this software.

Worldwide there is a huge experimental effort to do just that. In Australia, the Centre for Quantum Computation and Communication Technology(CQC2T) is a world-leader in two of the most promising types of hardware for a quantum computer: photons (particles of light), and the electrons of phosphorus atoms embedded in silicon.

Ultrasecure quantum communication

The other principal application of quantum technology is to make a message secure against any sort of hacker, no matter how powerful their computer is – even a quantum computer!

The basic idea is simple. Heisenberg's Uncertainty Principle implies that if you find out one property of a particle you necessarily create uncertainty in other properties. That is, quantum particles are disturbed by measurements. Because of this, an eavesdropper trying to read a secret message encoded on photons can be detected before they have a chance to obtain any real information.

Quantum communication protocols were first developed in the 1980s, and there are short-range systems in commercial operation in many countries, including an Australian one developed by CQC2T researchers.

The grand challenge, which is being tackled worldwide, including in CQC2T, is to extend the range of the communication. Because quantum messages cannot be copied, this requires other weird quantum effects, such as quantum teleportation.

The ultimate goal is to network quantum computers all over the globe, creating a global quantum internet.

As of now, we can only speculate about the scientific and other uses we will find for that, should it become a reality.

3. How QR codes work and what makes them dangerous – a computer scientist explains

Published: April 7, 2022 1.26pm BST

Scott Ruoti, Assistant Professor of Computer Science, University of Tennessee, USA

Among the many changes brought about by the pandemic is the widespread use of QR codes, graphical representations of digital data that can be printed and later scanned by a smartphone or other device.

QR codes have a wide range of uses that help people avoid contact with objects and close interactions with other people, including for sharing restaurant menus, email list sign-ups, car and home sales information, and checking in and out of medical and professional appointments.

QR codes are a close cousin of the bar codes on product packaging that cashiers scan with infrared scanners to let the checkout computer know what products are being purchased.

Bar codes store information along one axis, horizontally. QR codes store information in both vertical and horizontal axes, which allows them to hold significantly more data. That extra amount of data is what makes QR codes so versatile.

Anatomy of a QR code

While it is easy for people to read Arabic numerals, it is hard for a computer. Bar codes encode alphanumeric data as a series of black and white lines of various widths. At the store, bar codes record the set of numbers that specify a product's ID. Critically, data stored in bar codes is redundant. Even if part of the bar code is destroyed or obscured, it is still possible for a device to read the product ID.

QR codes are designed to be scanned using a camera, such as those found on your smartphone. QR code scanning is built into many camera apps for Android and iOS. QR codes are most often used to store web links; however, they can store arbitrary data, such as text or images.

When you scan a QR code, the QR reader in your phone's camera deciphers the code, and the resulting information triggers an action on your phone. If the QR code holds a URL, your phone will present you with the URL. Tap it, and your phone's default browser will open the webpage.

QR codes are composed of several parts: data, position markers, quiet zone and optional logos.



The QR code anatomy: data (1), position markers (2), quiet zone (3) and optional logos (4). Scott Ruoti, CC BY-ND

The data in a QR code is a series of dots in a square grid. Each dot represents a one and each blank a zero in binary code, and the patterns encode sets of numbers, letters or both, including URLs. At its smallest this grid is 21 rows by 21 columns, and at its largest it is 177 rows by 177 columns. In most cases, QR codes use black squares on a white background, making the dots easy to distinguish. However, this is not a strict requirement, and QR codes can use any color or shape for the dots and background.

Position markers are squares placed in a QR code's top-left, top-right, and bottom-left corners. These markers let a smartphone camera or other device orient the QR code when scanning it. QR codes are surrounded by blank space, the quiet zone, to help the computer determine where the QR code begins and ends. QR codes can include an optional logo in the middle.

Like barcodes, QR codes are designed with data redundancy. Even if as much as 30% of the QR code is destroyed or difficult to read, the data can still be recovered. In fact, logos are not actually part of the QR code; they cover up some of the QR code's data. However, due to the

QR code's redundancy, the data represented by these missing dots can be recovered by looking at the remaining visible dots.

Are QR codes dangerous?

QR codes are not inherently dangerous. They are simply a way to store data. However, just as it can be hazardous to click links in emails, visiting URLs stored in QR codes can also be risky in several ways.

The QR code's URL can take you to a phishing website that tries to trick you into entering your username or password for another website. The URL could take you to a legitimate website and trick that website into doing something harmful, such as giving an attacker access to your account. While such an attack requires a flaw in the website you are visiting, such vulnerabilities are common on the internet. The URL can take you to a malicious website that tricks another website you are logged into on the same device to take an unauthorized action.

A malicious URL could open an application on your device and cause it to take some action. Maybe you've seen this behavior when you clicked a Zoom link, and the Zoom application opened and automatically joined a meeting. While such behavior is ordinarily benign, an attacker could use this to trick some apps into revealing your data.

It is critical that when you open a link in a QR code, you ensure that the URL is safe and comes from a trusted source. Just because the QR code has a logo you recognize doesn't mean you should click on the URL it contains.

There is also a slight chance that the app used to scan the QR code could contain a vulnerability that allows malicious QR codes to take over your device. This attack would succeed by just scanning the QR code, even if you don't click the link stored in it. To avoid this threat, you should use trusted apps provided by the device manufacturer to scan QR codes and avoid downloading custom QR code apps.

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MEASUREMENT & CONVERSION

INTRODUCTION & EXERCISES

WHAT IS THE STANDARD
UNIT OF *length*?

Length IS GIVEN IN *meters*.
Length IS MEASURED IN *meters*.

c.g.s. system
f.p.s. system =

m.k.s. system **SI (International System of Units)**
The British system of physical units derived from the three fundamental units of length, mass, and time. It is now replaced, for scientific purposes, by SI units.

QUANTITY	ABBREVIATION	UNIT
length	<i>m</i> or <i>ft</i>	<i>meters (AE)/meters (BE)</i> or <i>feet</i>
mass		
	s	
force		
	Pa or bar	
volume		
density		
	J	
	W	
area		
flow (of a liquid)		
velocity		
	Hz	
	Ω	
conductance		
voltage		
	$^{\circ}\text{C}$ or $^{\circ}\text{F}$ or K	
electric current		

What does " " stand for?

In the equation $V = I R$, what does V stand for? What does R stand for? What does I stand for?

READING TEXT & TEXTWORK

Read the following text¹ and decide whether the statements below are true or false.

(1) We measure constantly. When we lift a pen, our senses measure its weight, and when we walk, our body registers acceleration, speed and the distance to our destination. In engineering we **conduct** tests to optimise our designs or check parameters to **verify** the quality of our products. But what exactly is a measurement?

(2) When measuring we quantify a certain property of something, e.g. a body; a substance or a physical **phenomenon**, by comparing it to a given standard unit. The result of the measurement therefore comes in two parts - a unit of measurement according to the property and a number for how many times the standard unit is contained in the measured object.

(3) Two of many properties of a steel **girder** are its mass and length. According to the Systeme International d'Unites, or SI units, the unit of the property mass is kilogram (kg) and the unit of the length is meter (m). [...] The SI system is a modern form of the metric system. It comprises a coherent system of units of measurement built around seven base units.

(4) If the measurement is taken by comparing the length of the girder to something with the same property, i.e. the length of a ruler is called a direct measurement, an indirect measurement, in contrast, looks at a different property, maybe the time sound takes to travel from one end of the girder to the other.

Every Measurement is Inexact

(5) No matter how **elaborate** a test setting might be, or how carefully we take a measurement, there's always a **margin of doubt**. Inaccuracy is an integral part of any measurement. Since a measurement is faulty by definition, several measurement attempts will return different results. The difference between the measured value and the true value is called an error. There are many sources of errors.

(6) Most importantly, all environmental effects should be taken into consideration. Temperature influences the length or volume of an object; other local conditions like air pressure, elevation or humidity might **affect** other properties. The measured item should also be stable, a shifting frequency or a **leaking** container are hard to measure.

(7) Naturally, the measuring instrument also plays an **important** role and any faults can easily be multiplied. Electric noise interfering with sensors or a worn out **vernier calliper** are just two of many examples. Equally important as the measuring instrument is the operator and the process he sets up. Reaction times in using a stopwatch vary from person to person and measuring the circumference of a cylinder with a folding rule is definitely poor practice.

¹ Adapted from: Engine, 02/2009. P. 17.

Exercise 1.

- A measurement consists of two parts. T / F
- The SI units are built on five base units. T / F
- As per the SI units centimeter is the unit of length. T / F
- Usually the circumference of a cylinder is measured with a folding rule. T / F

Exercise 2. Match these words with the words and expressions in bold which have the same or similar meaning.

- | | | | |
|-------------------------------------|-------|-------------------------------|-------|
| 1. carry out | | 2. vital | |
| 3. an observable event | | 4. to test | |
| 5. detailed | | 6. a give or take | |
| 7. measuring device | | 8. might influence | |
| 9. horizontal support in a building | | 10. to escape or pass through | |

Exercise 3. Label the illustration below which shows a vernier calliper. Use the words in the box.

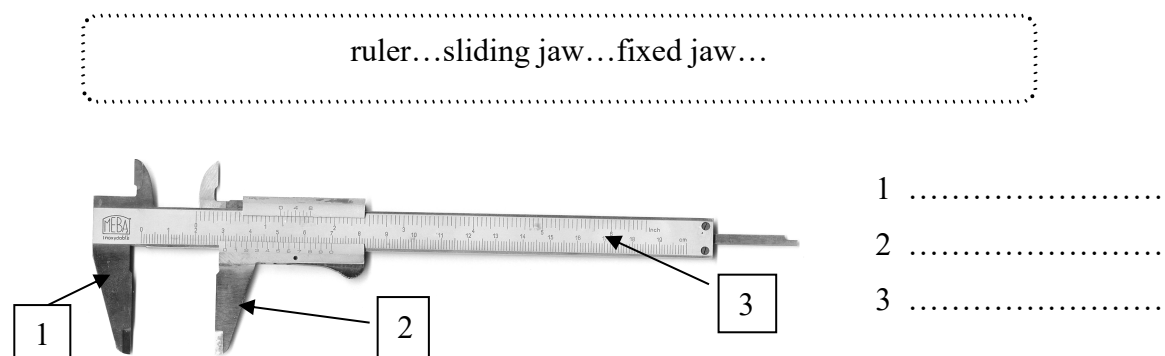


Fig. 1: A vernier calliper is an instrument for making accurate measurements of the thickness and diameter of mechanical parts.

Exercise 4. The International System of Units (SI units) is the modern form of the metric system. Complete the table of the seven base SI units below.

Quantity name	Unit symbol	Unit name
.....	m	meter (A.E.) / metre (B.E.)
mass		kilogram
.....	s	second
.....	A	ampere

thermodynamic temperature		kelvin
.....	mol	mole
luminous intensity	cd	

Exercise 5.

Before SI units were widely adopted around the world as the modern form of the metric system, the British system of English units and later imperial units were used in Britain, the Commonwealth and the US.

Look at the conversion table² below and decide whether these statements are true or false.

- | | |
|--|-------|
| 1. One mile is longer than one kilometer. | T / F |
| 2. The length of your little finger is about two inches. | T / F |
| 3. One pound (lb) is more than 0.5 kg | T / F |
| 4. In the US fuel is sold in gallons. | T / F |
| 5. A person who is 6'5" is taller than 1.78m | T / F |

Length	
1 inch	2.53 centimetres
0.394 inch	1 centimetre
1 foot = 12 inches	0.305 metre
1 yard = 3 feet	0.914 metre
1.094 yards	1 metre
1 mile = 1760 yards	1.609 kilometres
0.621 mile	1 kilometre
Volume Imperial	
1 fluid ounce (fl oz)	28.413 millilitres
1 pint (pt) = 20 fl oz	0.568 litre
1 gallon = 8 pts	4.546 litres

Volume US	
1 fluid ounce (fl oz)	29.574 millilitre
1 pint (pt) = 16 fl oz	0.473 litre
1 gallon = 8 pts	3.785 litres
Mass	
1 ounce (oz)	28.35 grams
0.035 ounce	1 gram
1 pound (lb) = 16 oz	0.454 kilograms
2.205 pounds	1 kilogram
1 stone = 14 lb	6.35 kilograms

² Engine, 02/2009. Pp. 16-17.

Exercise 6. Imperial Units & Conversion.

Lengths are commonly given in such units as inches, feet, yards, and miles.

1. 1 " $\approx$ _____ cm
2. 12 " = _____ ' and 3' = _____ yd
3. 1 mi = _____ km

Speed is given in miles per hour.

1. 100 km/h = _____ mph
2. What's the recommended speed limit on German freeways? _____ mph (130 km/h).

Liquids such as oil, gasoline, and milk are sold in pints, quarts, and gallons.

1. 1 gal = _____ qt and 1 qt = _____ pt
2. 1 US gal $\approx$ _____ l but 1 Imperial gallon $\approx$ _____ l

Weight is given in ounces or pounds.

1. 1 oz $\approx$ _____ g
2. 1 lb = _____ oz
3. 1 lb = _____ g
4. 1 kg = _____ lb

Area is given in square miles (large areas of land), acres (smaller land areas), or square feet (homes and offices).

$$1 \text{ square mile} \approx 2.6 \text{ km}^2$$

$$1 \text{ acre} \approx 4000 \text{ m}^2$$

$$1 \text{ square foot} = 0.09 \text{ m}^2$$

1. How large is Germany?
– Approximately _____ sq mi. (360 000 km²)
2. Land is very expensive in the Silicon Valley, so most single-family homes are on about one quarter to one half acre of land. How much is that in square meters?
– Approximately _____ m²
3. An average German home is about 100 to 200 m². How much is that in square feet?
– Approximately _____ ft²

Temperature is usually given in degrees Fahrenheit.

<i>under standard ambient conditions</i>	°C	°F (exactly)	°F (approximately)
boiling point of water	100 °C	=	$\approx$
normal body temperature	37 °C	=	$\approx$
standard room temperature	20 °C	=	$\approx$
freezing point of water	0 °C	=	$\approx$

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$$

$$^{\circ}\text{F} = 32 + ^{\circ}\text{C} \times 9/5$$